| | | |
|---|---|---|
| 7(a) | lepton(s) | **B1** |
| 7(b) | protons: 7 **and** neutrons: 6 | **A1** |
| 7(c) | $E = ½mv^2$ | **C1** |
| | $= 0.80 \times 10^6 \times 1.60 \times 10^{-19}$ | **C1** |
| | $= 1.28 \times 10^{-13}$ (J)<br>$v^2 = 2 \times 1.28 \times 10^{-13} / 2.2 \times 10^{-26}$<br>$v = 3.4 \times 10^6\,\mathrm{m\,s^{-1}}$ | **A1** |
| 7(d) | an (electron) neutrino/$\nu_{(e)}$ is also produced (and this has energy) | **B1** |